

PRAKTEK TEKNOLOGI IOT II
LAPORAN PRAKTEK KELOMPOK
TEKNOLOGI IOT II

Disusun dalam rangka memenuhi salah satu syarat kelengkapan nilai Praktikum
Mata Kuliah Teknologi IOT II



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KELAS : 3A
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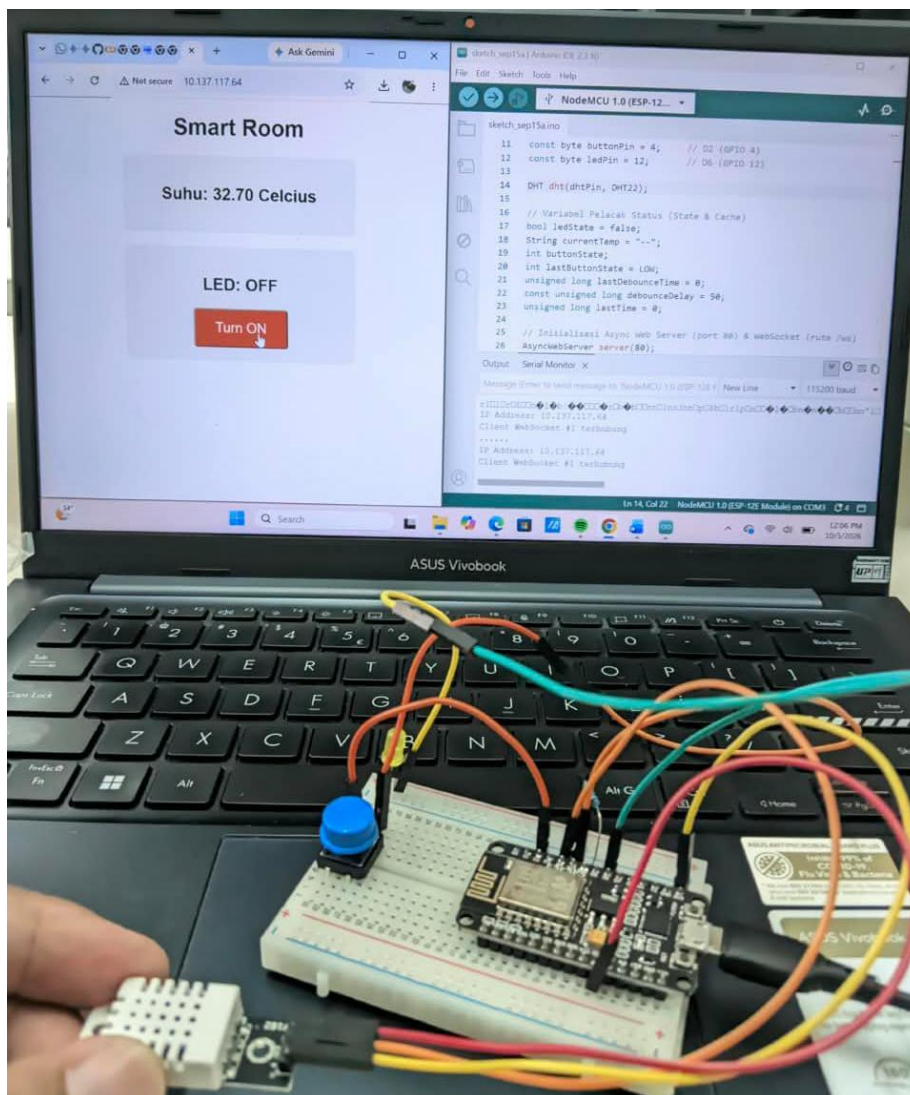
MENTRI PENDIDIKAN DAN KEBUDAYAAN PROVINSI LAMPUNG
POLITEKNIK NEGERI LAMPUNG
TAHUN AJARAN 2025 / 2026

Modul 4: Komunikasi Asinkron dan WebSockets

4.5 Praktikum

4.5.1 Praktikum 1: *Real-Time Dashboard & Sinkronisasi Tombol Fisik*

Skenario: Anda memiliki lampu LED (simulasi aktuator/lampu cerdas) yang bisa dihidupkan melalui sakelar fisik (*push button*) di *breadboard*, ATAU melalui tombol di antarmuka web *smartphone* Anda. Keduanya harus tersinkronisasi dua arah secara *real-time*. Jika Anda menekan tombol fisik di meja praktikum, status visual indikator di antarmuka *smartphone* harus otomatis berubah seketika, begitu pula sebaliknya jika dikontrol dari web.



Modul 5: Topologi Jaringan Terdesentralisasi (ESP-NOW dan Mesh)

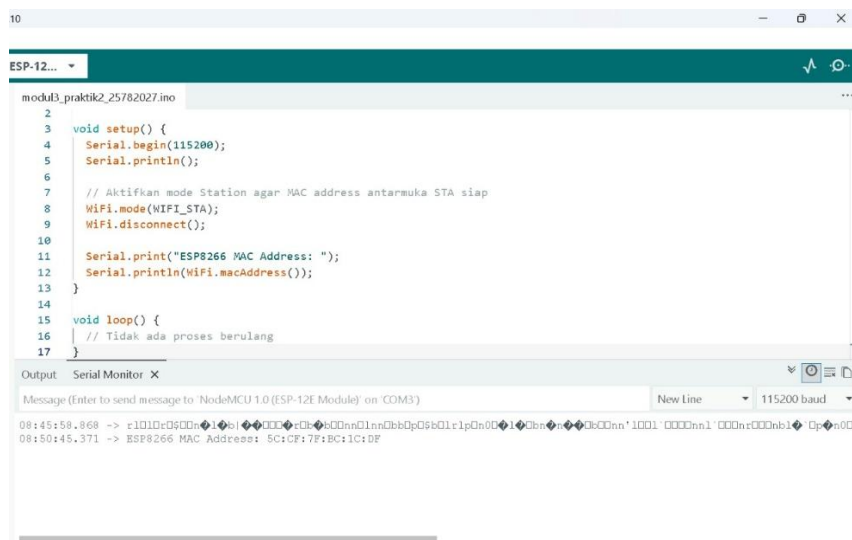
5.5 Praktikum

5.5.1 Praktikum 1: Identifikasi MAC Address Perangkat NodeMCU

Sebelum membangun jaringan ESP-NOW, setiap anggota kelompok wajib membaca dan mencatat alamat fisik (*MAC Address*) dari masing-masing *board* NodeMCU.

1. Tuliskan kode program pembaca MAC Address:

- NodeMCU 1:



```
modul3_praktik2_25782027.ino
2
3 void setup() {
4   Serial.begin(115200);
5   Serial.println();
6
7   // Aktifkan mode Station agar MAC address antarmuka STA siap
8   WiFi.mode(WIFI_STA);
9   WiFi.disconnect();
10
11   Serial.print("ESP8266 MAC Address: ");
12   Serial.println(WiFi.macAddress());
13 }
14
15 void loop() {
16   // Tidak ada proses berulang
17 }
```

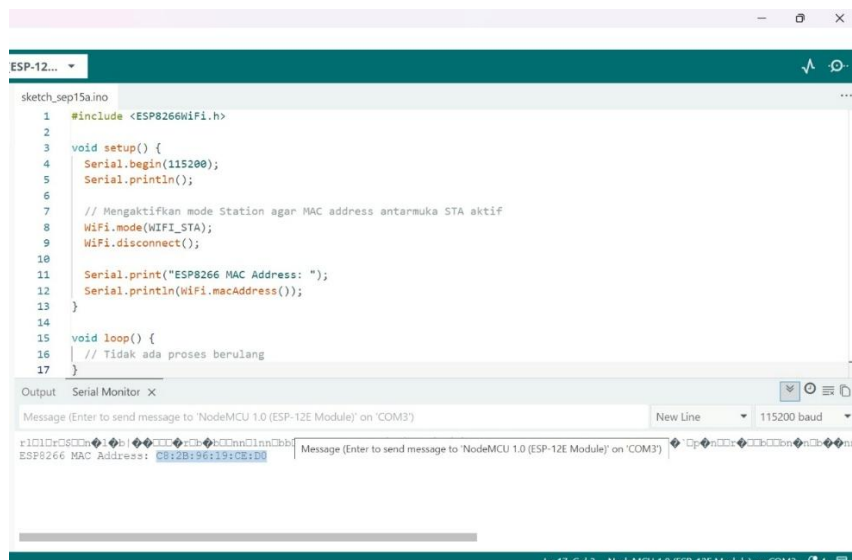
Output Serial Monitor X

Message (Enter to send message to 'NodeMCU 1.0 (ESP-12E Module)' on 'COM3')

New Line 115200 baud

08:45:59.868 -> ESP8266 MAC Address: 5C:C7:7F:8C:1C:D8

- NodeMCU 2:



```
sketch_sep15a.ino
1 #include <ESP8266WiFi.h>
2
3 void setup() {
4   Serial.begin(115200);
5   Serial.println();
6
7   // Mengaktifkan mode Station agar MAC address antarmuka STA aktif
8   WiFi.mode(WIFI_STA);
9   WiFi.disconnect();
10
11   Serial.print("ESP8266 MAC Address: ");
12   Serial.println(WiFi.macAddress());
13 }
14
15 void loop() {
16   // Tidak ada proses berulang
17 }
```

Output Serial Monitor X

Message (Enter to send message to 'NodeMCU 1.0 (ESP-12E Module)' on 'COM3')

New Line 115200 baud

ESP8266 MAC Address: C8:2B:96:19:CE:D0

- **NodeMCU 3:**

```

sketch_sep21aino
36  wifi.disconnect();
37
38  if (esp_now_init() != 0) {
39    Serial.println("Inisialisasi ESP-NOW Gagal!");
40    return;
41  }
42
43  // Tetapkan peran board sebagai Slave
44  esp_now_set_self_role(ESP_NOW_ROLE_SLAVE);
45  esp_now_register_recv_cb(OnDataRecv);
46
47  Serial.println("Receiver ESP-NOW Menunggu Paket...");
48
49
50 void loop() {
51   // Loop kosong: penerimaan data sepenuhnya ditangani oleh interrupt callback
52 }

```

Output Serial Monitor X

Message (Enter to send message to 'NodeMCU 1.0 (ESP-12E Module)' on 'COM9')

ESP8266 MAC Address: 50:02:91:41:22:F2

Paket masuk dari: 5c:cf:7f:bc:1c:df

Ukuran Data: 8 byte

Perintah ID: 101

Nilai Parameter: 74

5.5.2 Praktikum 2: Komunikasi Nirkabel One-to-Many (ESP-NOW)

Skenario: Satu unit NodeMCU bertindak sebagai **Controller / Sender**, menyiarkan paket data instruksi yang sama secara langsung ke dua unit NodeMCU lain yang bertindak sebagai **Slave / Receiver** tanpa perantara router.

1. Pemrograman Sisi Pengirim (Controller):

```

modul3_praktik2_25782027.ino
1  #include <ESP8266WiFi.h>
2  #include <espnow.h>
3
4  typedef struct struct_sensor {
5    int id;
6    float pembacaan1;
7    float pembacaan2;
8  } struct_sensor;
9
10 struct_sensor dataTerima;
11
12 // Array penampung data terakhir: Index 0 untuk Node 1, Index 1 untuk Node 2
13 struct_sensor records[2];
14
15 void OnDataRecv(uint8_t *mac_addr, uint8_t *incomingData, uint8_t len) {
16   memcpy(&dataTerima, incomingData, sizeof(dataTerima));

```

Output Serial Monitor X

Message (Enter to send message to 'NodeMCU 1.0 (ESP-12E Module)' on 'COM3')

09:16:36.884 -> Parameter 1 : 25.80

09:16:36.884 -> Parameter 2 : 52.60

09:16:36.884 -> =====

09:16:36.978 -> DATA TERBARU DARI NODE #2

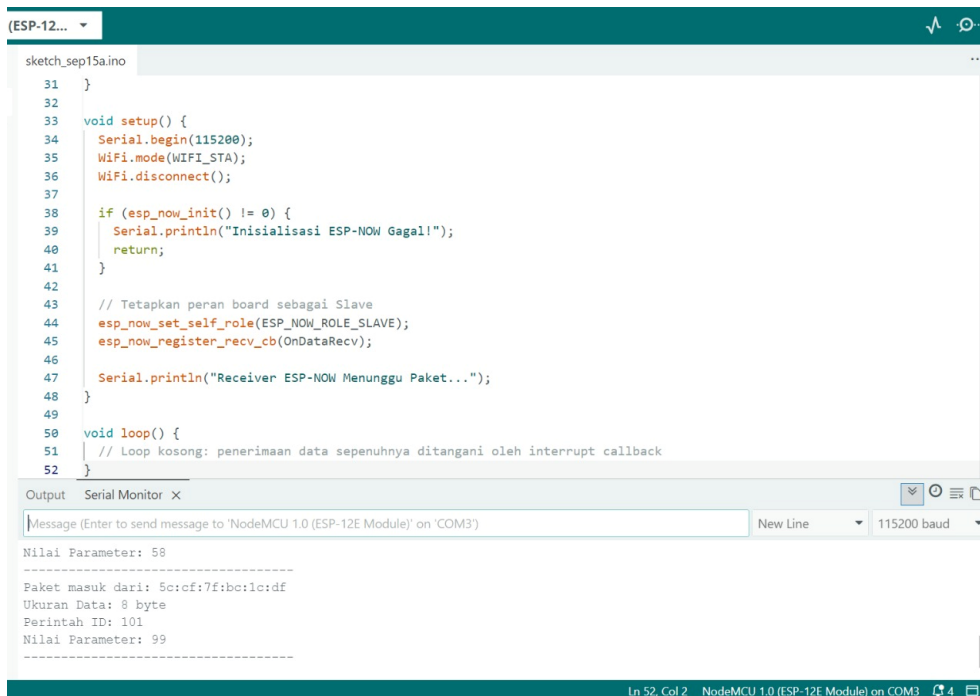
09:16:36.978 -> Parameter 1 : 32.30

09:16:36.978 -> Parameter 2 : 66.30

09:16:36.978 -> =====

3. Pemrograman Sisi Penerima (Slave):

NodeMCU 2:



The screenshot shows the Arduino IDE interface. The top bar indicates the board is "ESP-12...". The code editor displays a sketch named "sketch_sep15a.ino" with the following code:

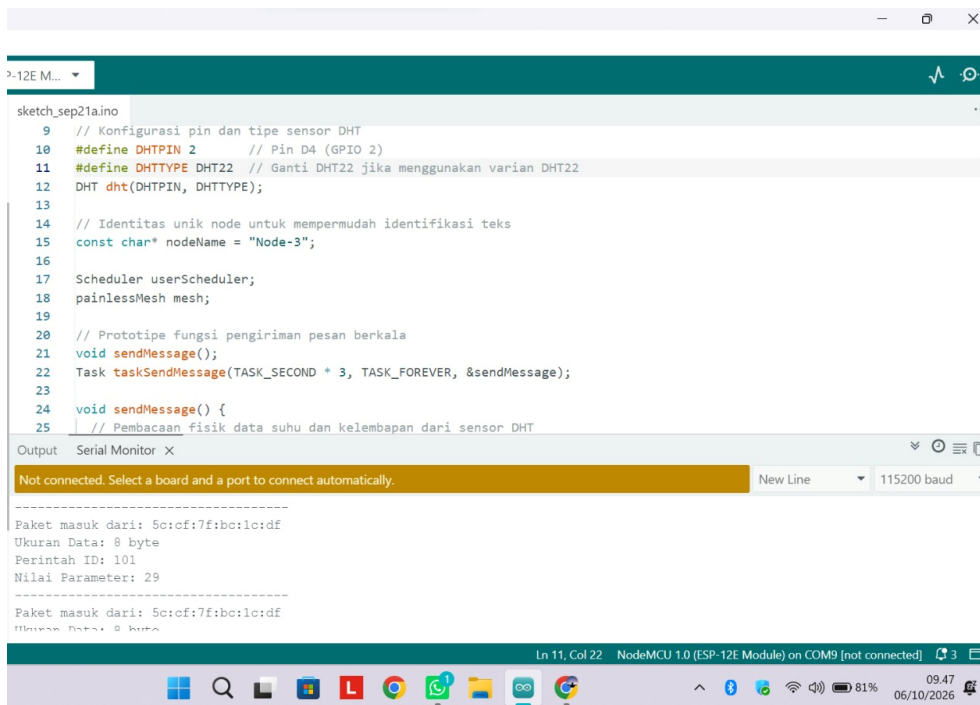
```
31 }
32
33 void setup() {
34   Serial.begin(115200);
35   Wifi.mode(WIFI_STA);
36   Wifi.disconnect();
37
38   if (esp_now_init() != 0) {
39     Serial.println("Inisialisasi ESP-NOW Gagal!");
40     return;
41   }
42
43   // Tetapkan peran board sebagai Slave
44   esp_now_set_self_role(ESP_NOW_ROLE_SLAVE);
45   esp_now_register_recv_cb(OnDataRecv);
46
47   Serial.println("Receiver ESP-NOW Menunggu Paket...");
48 }
49
50 void loop() {
51   // Loop kosong: penerimaan data sepenuhnya ditangani oleh interrupt callback
52 }
```

The Serial Monitor is open, showing the following output:

```
Nilai Parameter: 58
-----
Paket masuk dari: 5c:cf:7f:bc:1c:df
Ukuran Data: 8 byte
Perintah ID: 101
Nilai Parameter: 99
-----
```

The status bar at the bottom indicates "Ln 52, Col 2 NodeMCU 1.0 (ESP-12E Module) on COM3".

NodeMCU 3:



The screenshot shows the Arduino IDE interface. The top bar indicates the board is "ESP-12E M...". The code editor displays a sketch named "sketch_sep21a.ino" with the following code:

```
9 // Konfigurasi pin dan tipe sensor DHT
10 #define DHTPIN 2 // Pin D4 (GPIO 2)
11 #define DHTTYPE DHT22 // Ganti DHT22 jika menggunakan varian DHT22
12 DHT dht(DHTPIN, DHTTYPE);
13
14 // Identitas unik node untuk mempermudah identifikasi teks
15 const char* nodeName = "Node-3";
16
17 Scheduler userScheduler;
18 painlessMesh mesh;
19
20 // Prototipe fungsi pengiriman pesan berkala
21 void sendMessage();
22 Task taskSendMessage(TASK_SECOND * 3, TASK_FOREVER, &sendMessage);
23
24 void sendMessage() {
25   // Pembacaan fisik data suhu dan kelembapan dari sensor DHT
```

The Serial Monitor is open, showing the following output:

```
Not connected. Select a board and a port to connect automatically.
-----
Paket masuk dari: 5c:cf:7f:bc:1c:df
Ukuran Data: 8 byte
Perintah ID: 101
Nilai Parameter: 29
-----
Paket masuk dari: 5c:cf:7f:bc:1c:df
Ukuran Data: 8 byte
```

The status bar at the bottom indicates "Ln 11, Col 22 NodeMCU 1.0 (ESP-12E Module) on COM9 [not connected]". The Windows taskbar is visible at the bottom of the screen.

5.5.3 Praktikum 3: Pengumpulan Data Telemetry Many-to-One (ESP-NOW)

Skenario: Dua node sensor di lapangan memancarkan data pengukuran masing-masing ke sebuah node konsentrator pusat (*Aggregator Hub / Base Station*).

Pemrograman Sisi Node Pengirim:

```
modul3_praktik2_25782027.ino
1  #include <ESP8266WiFi.h>
2  #include <espnow.h>
3
4  // GANTI dengan MAC Address Board Penerima 1 dan Board Penerima 2 Andai
5  uint8_t receiver1[] = {0xC8, 0x2B, 0x96, 0x19, 0xCE, 0xD0};
6  uint8_t receiver2[] = {0x50, 0x02, 0x91, 0x41, 0x22, 0xF2};
7
8  // Definisi struktur data paket instruksi
9  typedef struct struct_pesan {
10     int perintahId;
11     int nilaiParameter;
12 } struct_pesan;
13
14 struct_pesan paketKirim;
15
16 unsigned long previousMillis = 0;
```

Output Serial Monitor X

Message (Enter to send message to 'NodeMCU 1.0 (ESP-12E Module)' on 'COM3')

New Line 115200 baud

```
09:04:27.687 -> Kirim paket ke: c8:2b:96:19:ce:d0 | Status: Berhasil Diterima
09:04:27.721 -> Kirim paket ke: 50:02:91:41:22:f2 | Status: Berhasil Diterima
09:04:29.731 -> Kirim paket ke: c8:2b:96:19:ce:d0 | Status: Berhasil Diterima
09:04:29.731 -> Kirim paket ke: 50:02:91:41:22:f2 | Status: Berhasil Diterima
09:04:31.688 -> Kirim paket ke: c8:2b:96:19:ce:d0 | Status: Berhasil Diterima
09:04:31.722 -> Kirim paket ke: 50:02:91:41:22:f2 | Status: Berhasil Diterima
09:04:33.687 -> Kirim paket ke: c8:2b:96:19:ce:d0 | Status: Berhasil Diterima
09:04:33.728 -> Kirim paket ke: 50:02:91:41:22:f2 | Status: Berhasil Diterima
```

Pemrograman Sisi Hub Konsentrator:

NodeMCU 2:

```
ESP-12...
sketch_sep15a.ino
1  #include <ESP8266WiFi.h>
2  #include <espnow.h>
3
4  // GANTI dengan MAC Address milik BOARD PENERIMA PUSAT (HUB)!
5  uint8_t hubMacAddress[] = {0x5C, 0xCF, 0x7F, 0xBC, 0x1C, 0xDF};
6
7  // Identitas unik node (Ubah angka ini untuk node yang berbeda!)
8  #define BOARD_ID 1
9
10 typedef struct struct_sensor {
11     int id;
12     float pembacaan1;
13     float pembacaan2;
14 } struct_sensor;
15
16 struct_sensor dataKirim;
17
18 unsigned long prevTime = 0;
19 const unsigned long sendInterval = 3000; // Kirim data setiap 3 detik
20
21 void OnDataSent(uint8_t *mac_addr, uint8_t sendStatus) {
22     Serial.print("Pengiriman Node #");
```

Output Serial Monitor X

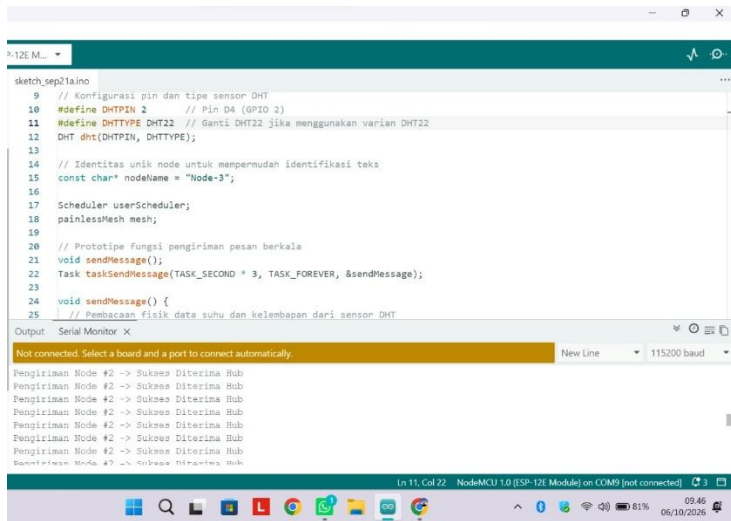
Message (Enter to send message to 'NodeMCU 1.0 (ESP-12E Module)' on 'COM3')

New Line 115200 baud

```
Pengiriman Node #1 -> Sukses Diterima Hub
Pengiriman Node #1 -> Sukses Diterima Hub
Pengiriman Node #1 -> Sukses Diterima Hub
Pengiriman Node #1 -> Sukses Diterima Hub
Pengiriman Node #1 -> Sukses Diterima Hub
Pengiriman Node #1 -> Sukses Diterima Hub
Pengiriman Node #1 -> Sukses Diterima Hub
Pengiriman Node #1 -> Sukses Diterima Hub
```

Ln 8, Col 19 NodeMCU 1.0 (ESP-12E Module) on COM3

NodeMCU 3:



```
sketch_sep21aino
9 // Konfigurasi pin dan tipe sensor DHT
10 #define DHTPIN 2 // Pin D4 (GPIO 2)
11 #define DHTTYPE DHT22 // Ganti DHT22 jika menggunakan varian DHT22
12 DHT dht(DHTPIN, DHTTYPE);
13
14 // Identitas unik node untuk mempermudah identifikasi teks
15 const char* nodeName = "Node-3";
16
17 Scheduler userScheduler;
18 painlessMesh mesh;
19
20 // Prototipe fungsi pengiriman pesan berkala
21 void sendMessage();
22 Task taskSendMessage(TASK_SECOND * 3, TASK_FOREVER, &sendMessage);
23
24 void sendMessage() {
25 // Pembacaan fisik data suhu dan kelembapan dari sensor DHT
```

Output Serial Monitor X

Not connected. Select a board and a port to connect automatically.

Pengiriman Node #2 -> Sukses Diterima Hub
Pengiriman Node #2 -> Sukses Diterima Hub
Pengiriman Node #2 -> Sukses Diterima Hub
Pengiriman Node #2 -> Sukses Diterima Hub
Pengiriman Node #2 -> Sukses Diterima Hub
Pengiriman Node #2 -> Sukses Diterima Hub
Pengiriman Node #2 -> Sukses Diterima Hub
Pengiriman Node #2 -> Sukses Diterima Hub
Pengiriman Node #2 -> Sukses Diterima Hub
Pengiriman Node #2 -> Sukses Diterima Hub

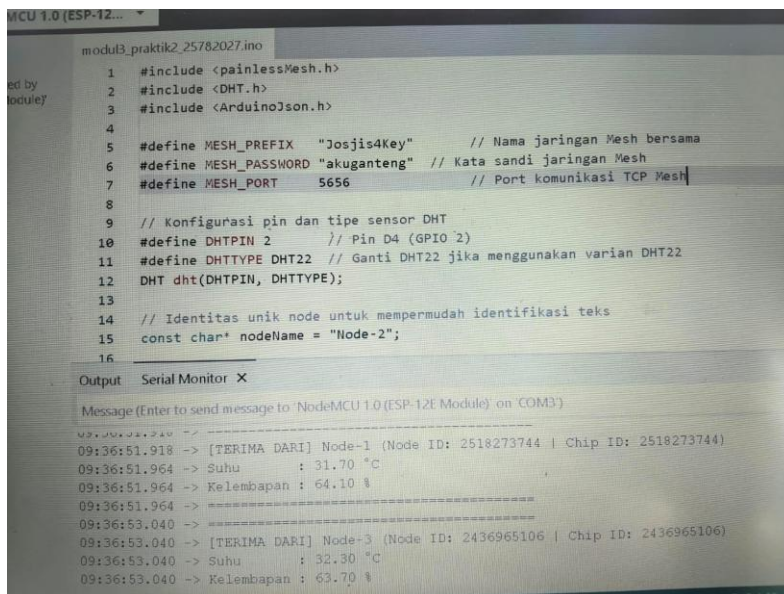
In 11, Col 22 NodeMCU 1.0 [ESP-12E Module] on COM9 [not connected]

5.5.4 Praktikum 4: Jaringan Sensor Ad-Hoc Multi-Hop (painlessMesh)

Skenario: Seluruh board dalam kelompok dihubungkan ke sensor suhu & kelembapan riil (DHT11/DHT22) dan diintegrasikan ke dalam sebuah jaringan rantai otonom (*Mesh*). Setiap board secara berkala membaca sensor fisiknya, mengemas data telemetri ke dalam format **JSON**, dan menyiarkannya ke seluruh anggota jaringan tanpa perlu mengetahui alamat MAC masing-masing.

Berikan nama pengenalan unik pada variabel nodeName di masing-masing board (misal: "Node-1", "Node-2", "Node-3"):

NodeMCU Utama:



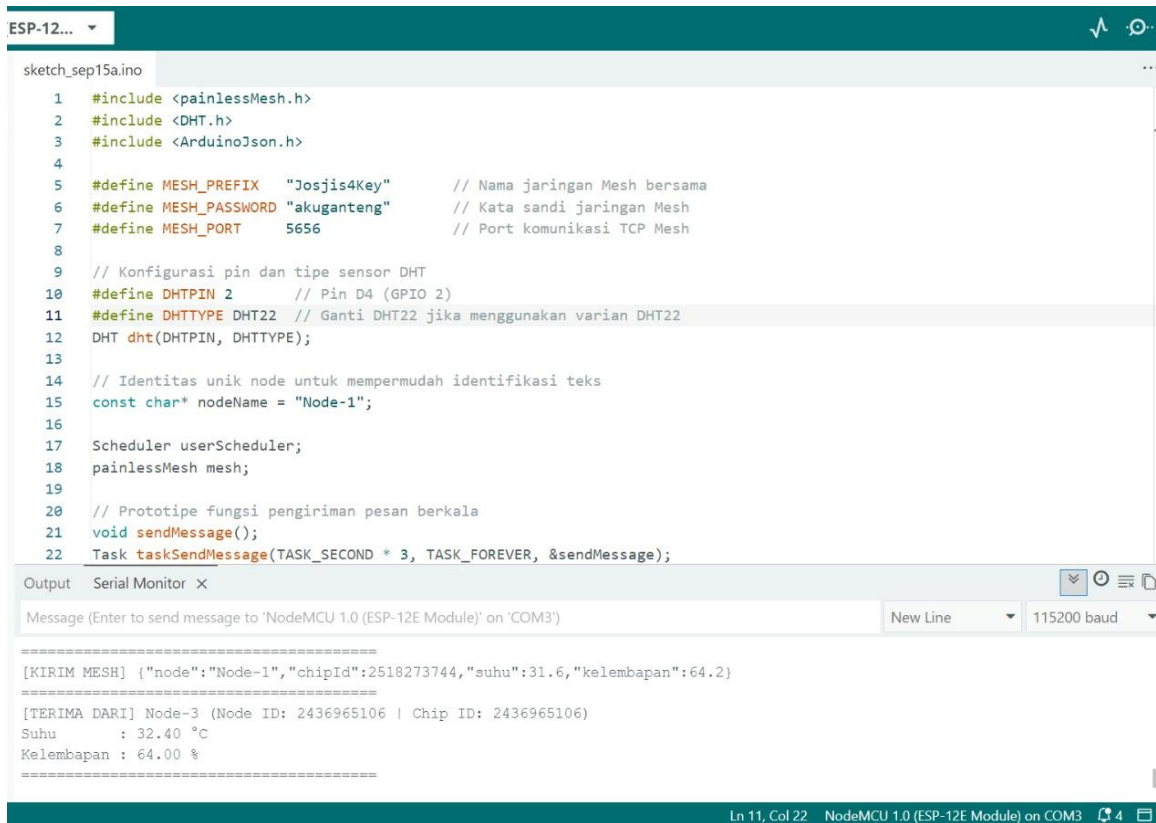
```
modul3_praktik2_25782027.ino
1 #include <painlessMesh.h>
2 #include <DHT.h>
3 #include <ArduinoJson.h>
4
5 #define MESH_PREFIX "Josjis4Key" // Nama jaringan Mesh bersama
6 #define MESH_PASSWORD "akuganteng" // Kata sandi jaringan Mesh
7 #define MESH_PORT 5656 // Port komunikasi TCP Mesh
8
9 // Konfigurasi pin dan tipe sensor DHT
10 #define DHTPIN 2 // Pin D4 (GPIO 2)
11 #define DHTTYPE DHT22 // Ganti DHT22 jika menggunakan varian DHT22
12 DHT dht(DHTPIN, DHTTYPE);
13
14 // Identitas unik node untuk mempermudah identifikasi teks
15 const char* nodeName = "Node-2";
16
```

Output Serial Monitor X

Message (Enter to send message to 'NodeMCU 1.0 (ESP-12E Module)' on COM3)

09:36:51.918 -> [TERIMA DARI] Node-1 (Node ID: 2518273744 | Chip ID: 2518273744)
09:36:51.964 -> Suhu : 31.70 °C
09:36:51.964 -> Kelembapan : 64.10 %
09:36:51.964 -> =====
09:36:53.040 -> [TERIMA DARI] Node-3 (Node ID: 2436965106 | Chip ID: 2436965106)
09:36:53.040 -> Suhu : 32.30 °C
09:36:53.040 -> Kelembapan : 63.70 %

NodeMCU 2:



ESP-12...

sketch_sep15a.ino

```
1 #include <painlessMesh.h>
2 #include <DHT.h>
3 #include <ArduinoJson.h>
4
5 #define MESH_PREFIX "Josjis4Key" // Nama jaringan Mesh bersama
6 #define MESH_PASSWORD "akuganteng" // Kata sandi jaringan Mesh
7 #define MESH_PORT 5656 // Port komunikasi TCP Mesh
8
9 // Konfigurasi pin dan tipe sensor DHT
10 #define DHTPIN 2 // Pin D4 (GPIO 2)
11 #define DHTTYPE DHT22 // Ganti DHT22 jika menggunakan varian DHT22
12 DHT dht(DHTPIN, DHTTYPE);
13
14 // Identitas unik node untuk mempermudah identifikasi teks
15 const char* nodeName = "Node-1";
16
17 Scheduler userScheduler;
18 painlessMesh mesh;
19
20 // Prototipe fungsi pengiriman pesan berkala
21 void sendMessage();
22 Task taskSendMessage(TASK_SECOND * 3, TASK_FOREVER, &sendMessage);
```

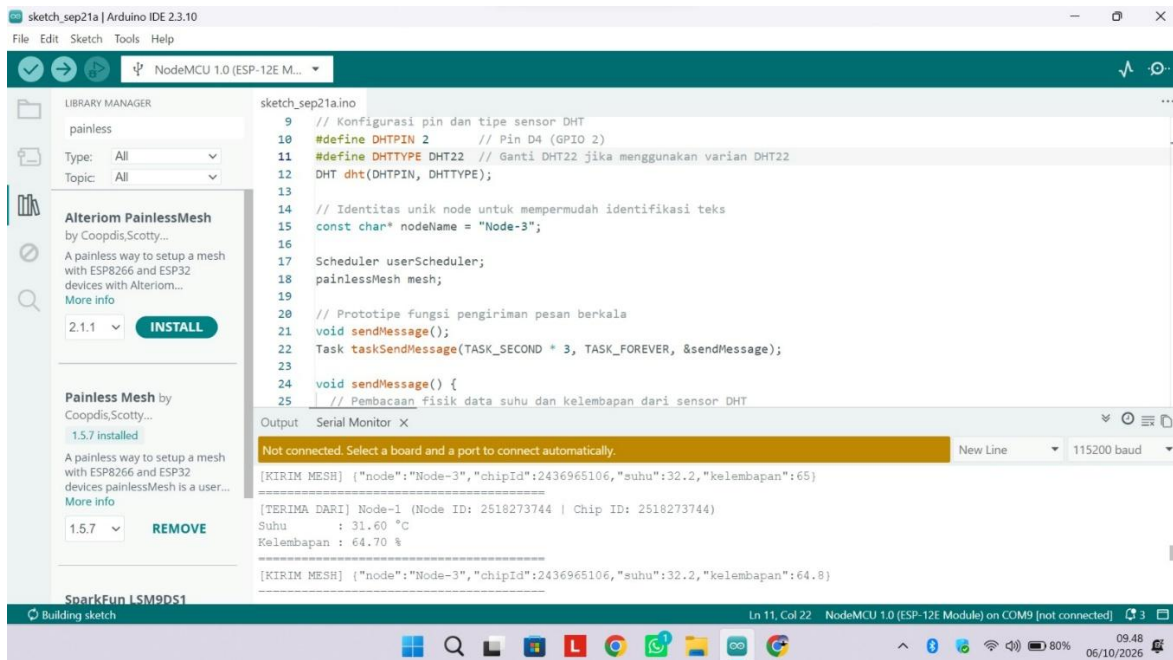
Output Serial Monitor X

Message (Enter to send message to 'NodeMCU 1.0 (ESP-12E Module)' on 'COM3') New Line 115200 baud

```
[KIRIM MESH] {"node":"Node-1","chipId":2518273744,"suhu":31.6,"kelembapan":64.2}
[TERIMA DARI] Node-3 (Node ID: 2436965106 | Chip ID: 2436965106)
Suhu : 32.40 °C
Kelembapan : 64.00 %
```

Ln 11, Col 22 NodeMCU 1.0 (ESP-12E Module) on COM3

NodeMCU 3:



sketch_sep21a | Arduino IDE 2.3.10

File Edit Sketch Tools Help

NodeMCU 1.0 (ESP-12E M...

LIBRARY MANAGER

painless

Type: All

Topic: All

Alteriom PainlessMesh by Coopdis,Scotty...

A painless way to setup a mesh with ESP8266 and ESP32 devices with Alteriom...

More info

2.1.1 INSTALL

Painless Mesh by Coopdis,Scotty...

1.5.7 installed

A painless way to setup a mesh with ESP8266 and ESP32 devices painlessMesh is a user...

More info

1.5.7 REMOVE

SparkFun LSM9DS1

Building sketch

sketch_sep21a.ino

```
9 // Konfigurasi pin dan tipe sensor DHT
10 #define DHTPIN 2 // Pin D4 (GPIO 2)
11 #define DHTTYPE DHT22 // Ganti DHT22 jika menggunakan varian DHT22
12 DHT dht(DHTPIN, DHTTYPE);
13
14 // Identitas unik node untuk mempermudah identifikasi teks
15 const char* nodeName = "Node-3";
16
17 Scheduler userScheduler;
18 painlessMesh mesh;
19
20 // Prototipe fungsi pengiriman pesan berkala
21 void sendMessage();
22 Task taskSendMessage(TASK_SECOND * 3, TASK_FOREVER, &sendMessage);
23
24 void sendMessage() {
25 // Pembacaan fisik data suhu dan kelembapan dari sensor DHT
```

Output Serial Monitor X

Not connected. Select a board and a port to connect automatically. New Line 115200 baud

```
[KIRIM MESH] {"node":"Node-3","chipId":2436965106,"suhu":32.2,"kelembapan":65}
[TERIMA DARI] Node-1 (Node ID: 2518273744 | Chip ID: 2518273744)
Suhu : 31.60 °C
Kelembapan : 64.70 %
[KIRIM MESH] {"node":"Node-3","chipId":2436965106,"suhu":32.2,"kelembapan":64.8}
```

Ln 11, Col 22 NodeMCU 1.0 (ESP-12E Module) on COM9 [not connected]

09.48 06/10/2026

FOTO HASIL ALAT PRAKTEK:

